

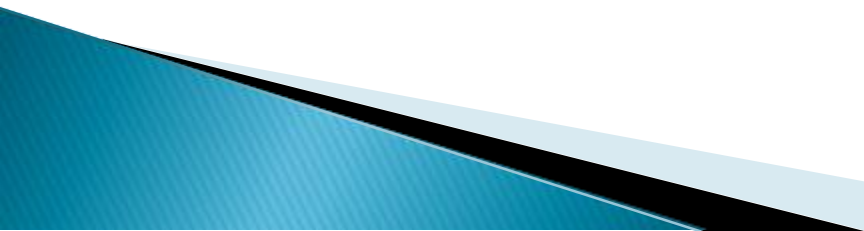
Dermatological Infections: Bacterial, Fungal, Viral

Georgia Allen, DO
GERD

Adapted from American Academy of Dermatology Cases

PART ONE: BACTERIA

Learning Objectives

- ▶ Describe the morphology of common cutaneous bacterial infections
 - ▶ Discuss the bacterial etiologies of cellulitis and erysipelas
 - ▶ Recognize clinical patterns and risk factors that suggest MRSA
 - ▶ Recommend initial steps for the evaluation and treatment of common cutaneous bacterial infections
 - ▶ Recognize characteristic features of necrotizing fasciitis and the need for emergent treatment, including surgical intervention
- 

PART TWO: FUNGUS

Learning Objectives

- ▶ Identify and describe the morphologies of superficial fungal infections
- ▶ Describe the correct procedure for performing a KOH examination and interpreting the results
- ▶ Recognize the use and limitations of KOH examination and fungal cultures to diagnose fungal infections
- ▶ Recommend an initial treatment plan for an adult with onychomycosis
- ▶ Recommend an initial treatment plan for an adult with tinea versicolor, candidal intertrigo, and seborrheic dermatitis

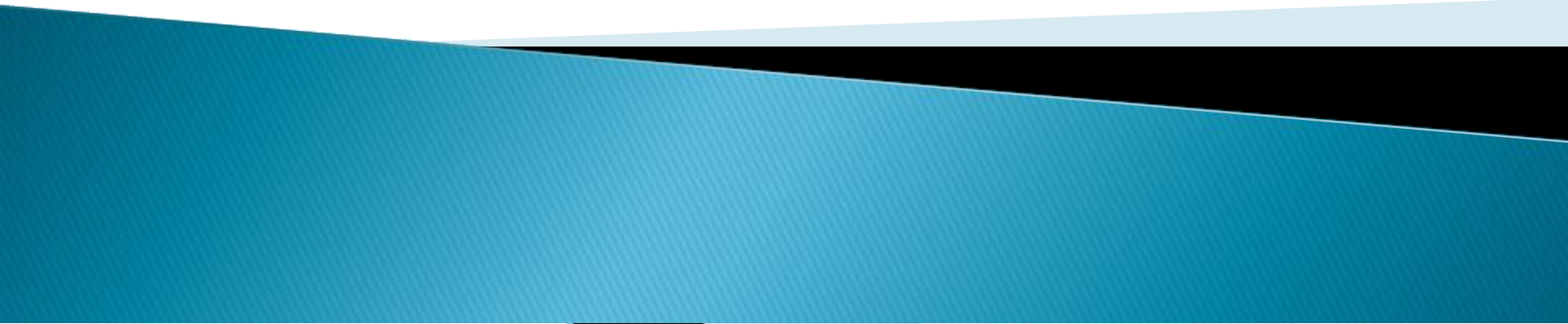
PART THREE: VIRUS

Learning Objectives

- ▶ Identify and describe the morphology of various types of warts
- ▶ Discuss the pathogenesis of warts
- ▶ Develop an initial treatment plan for a patient with warts
- ▶ Determine when vaccines can be helpful for wart management

Neal Tolson

Case



History

- ▶ HPI: Mr. Tolson is a 55-year-old man who presents with 5 days of worsening right lower extremity pain and a red rash. He reports recent fevers and chills since he returned from a camping trip last week.
- ▶ PMH: arthritis
- ▶ Medications: occasional NSAIDs, multivitamin
- ▶ Allergies: no known drug allergies
- ▶ Family history: father with history of melanoma
- ▶ Social history: lives in the city with his wife, two grown children
- ▶ Health-related behaviors: no alcohol, tobacco or drug use
- ▶ ROS: able to bear weight, no itching

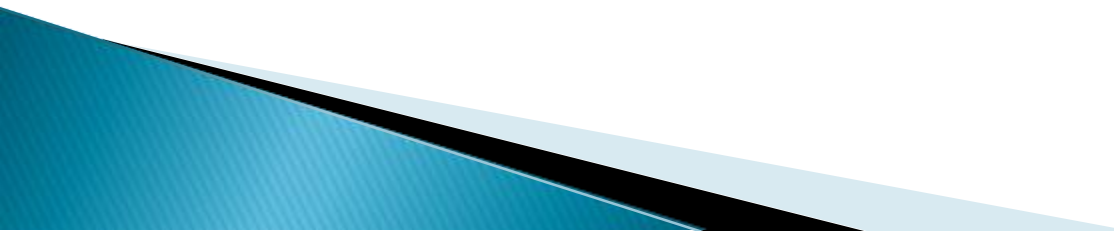


Physical Exam

- ▶ Vital signs: T 100.4, HR 90, BP 120/70, RR 14, O2 sat 97% on RA
- ▶ Skin: erythematous plaque with ill-defined borders over the right medial malleolus. Lesion is tender to palpation. Lymphatic streaking is noted (not shown).
- ▶ Tender, slightly enlarged right inguinal lymph nodes (not shown)
- ▶ Laboratory data: WBC 12,000 (75% neutrophils, 10% bands), Hct 44, Plts 335



What is the most likely diagnosis?

- a) Bacterial folliculitis
 - b) Cellulitis
 - c) Necrotizing fasciitis
 - d) Stasis dermatitis
 - e) Tinea corporis
- 

Answer...



- a) Bacterial folliculitis – would expect pustules and papules centered on hair follicles, without systemic signs of infection
- b) **Cellulitis**
- c) Necrotizing fasciitis -- Would expect rapidly expanding rash, usually appears as a dusky, edematous, red plaque.
- d) Stasis dermatitis -- Although found in similar location, stasis dermatitis often presents with pruritus and scale, which may erode or crust. Wouldn't expect fever or elevated wbc
- e) Tinea corporis –would expect annular plaque with elevated border and central clearing. Painless, without fever or elevated



Diagnosis: Cellulitis

- ▶ Cellulitis is c
- ▶ **Portal of entr**
- (e.g., tinea p
- ▶ Spreading er
- tender plaqu
- ▶ More commo
- ▶ **Lymphangitic**
- nodes



(year)
ion

lymph

Cellulitis: Risk Factors

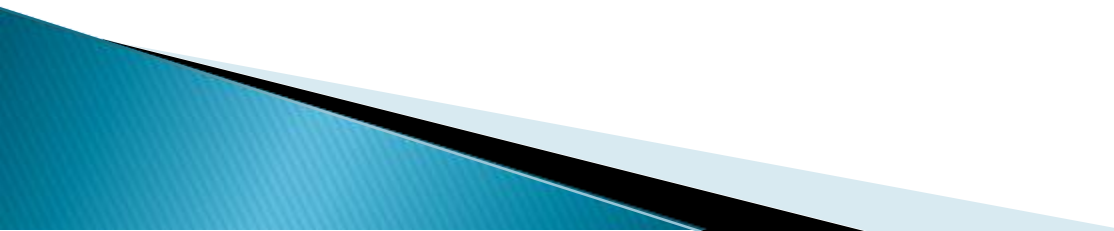
- ▶ Local trauma (bug bite, laceration, abrasion, puncture wound)
- ▶ Underlying skin lesion (furuncle, ulcer)
- ▶ Inflammation (local dermatitis, radiation therapy)
- ▶ Edema and impaired lymphatics in the affected area
- ▶ Preexisting skin infection (impetigo, tinea pedis)

Cellulitis: Etiology



- ▶ 80% of cases are caused by gram positive organisms
- ▶ Group A streptococcus and Staphylococcus aureus are the most common causal pathogens
- ▶ Think of other organisms if there have been unusual exposures:
 - Animal bite: *Pasteurella multocida*
 - Human bite: *Eikenella corrodens*

What is the next best step in management?

- a) Apply topical antibiotics
 - b) Apply topical steroids, compression wraps, and encourage leg elevation
 - c) Begin antibiotics immediately with coverage for gram positive bacteria
 - d) Order an imaging study
- 

Answer...

- a) Apply topical antibiotics (not effective)
- b) Apply topical steroids, compression wraps, and encourage leg elevation (this is the treatment for stasis dermatitis, not cellulitis)
- c) **Begin antibiotics immediately with coverage for gram positive bacteria**
- d) Order an imaging study (radiographic examination is not necessary for routine evaluation of patients with cellulitis)



Cellulitis: Treatment

- ▶ Untreated cellulitis may lead to sepsis and death
- ▶ Empirically treat for β -hemolytic streptococci (group A streptococcus)
- ▶ Empirically treat for **MRSA** if purulent drainage or exudate with no drainable abscess, hospitalized patients, drug users
- ▶ **Elevation** of the involved area
- ▶ Treat tinea pedis if present
- ▶ **Culture what you can!** (abscesses, purulence)
- ▶ Revise tx PRN

Antibiotics Used to Treat MRSA

Drug	Dosage (adult dosing with normal renal function)	Comments
Clindamycin	600 mg/kg IV Q8H 300-450 mg PO TID	Excellent tissue and abscess penetration. Risk for C. difficile Inducible resistance in MRSA
Trimethoprim-Sulfamethoxazole (TMP/SMX)	1 or 2 double-strength tablets PO BID	Unreliable for S. pyogenes (will need to combine with amoxicillin to cover for group A strep)
Doxycycline	100 mg PO BID	Unreliable for S. pyogenes (will need to combine with amoxicillin to cover for group A strep). Do not use in children < 8 years old.
Linezolid	600 mg IV Q12H 600 mg PO BID	Expensive. No cross-resistance with other antibiotic classes
Vancomycin	1g IV Q12H	Parenteral drug of choice for treatment of infections caused by MRSA

Chase DeGeorgio

Does he have cellulitis?

1. Yes
2. No



Yes– a type of cellulitis called Erysipelas



Erysipelas



- ▶ Erysipelas is a superficial cellulitis with marked dermal lymphatic involvement (causing the skin to be edematous or raised)
 - Main pathogen is group A streptococcus
- ▶ Usually affects the lower extremities and the face
- ▶ Presents with pain, superficial erythema, and plaque-like edema with a sharply defined margin to normal tissue
- ▶ May be associated with a high white count ($>20,000/\text{mCL}$)
- ▶ May be preceded by chills, fever, headache, vomiting, and joint pain

Example of Erysipelas



Large, shiny
erythematous plaque
with sharply demarcated
borders located on the
leg

What is the most appropriate treatment?

- a) Oral antibiotics
- b) Oral steroids
- c) Topical antibiotics
- d) Topical moisturizers
- e) Topical steroids



Answer...



a) **Oral antibiotics** are the most appropriate therapy in uncomplicated erysipelas.

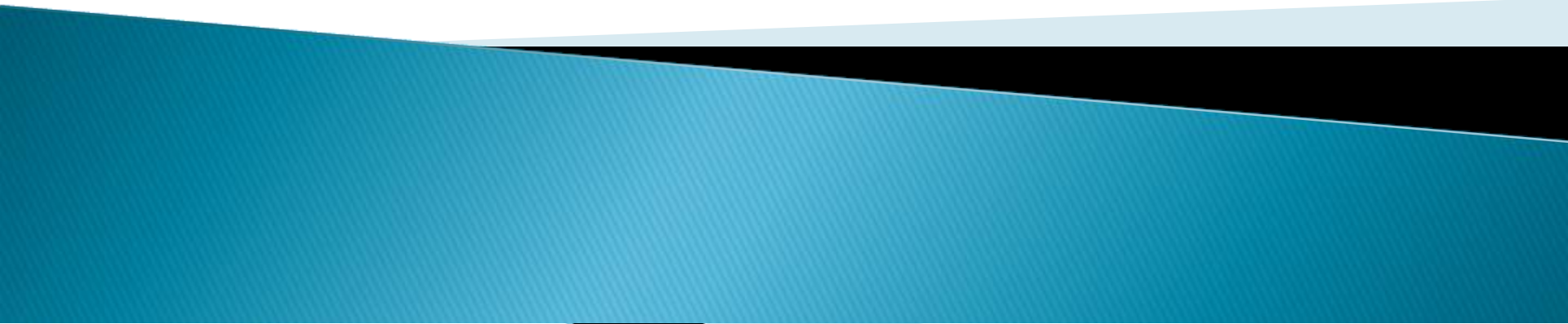
Erysipelas: Treatment



- ▶ Immediate empiric antibiotic therapy should be started (**cover most common pathogen – Streptococcus**)
- ▶ Monitor patients closely and revise therapy if there is a poor response to initial treatment
- ▶ Elevation of the involved area
- ▶ Treat tinea pedis if present

Jesse Hammel

Case



History

- ▶ HPI: Mr. Hammel is a 27-year-old man with a history of “skin popping” (subcutaneous or intradermal injection of drug) who presents to the emergency department with a painful, enlarging mass on his right arm for the last two days.
- ▶ PMH: History of skin and soft tissue infections, hospitalized with MRSA bacteremia two years ago
- ▶ Medications: none
- ▶ Allergies: no known drug allergies
- ▶ Family history: unremarkable
- ▶ Social history: lives with friends in an apartment, works in retail
- ▶ Health-related behaviors: IVDU (intravenous drug use), including skin popping. No tobacco or alcohol use.
- ▶ ROS: no fevers, sweats or chills

Skin Exam

- ▶ Erythematous, warm, fluctuant nodule with several small pustules throughout the surface
- ▶ Very tender to palpation

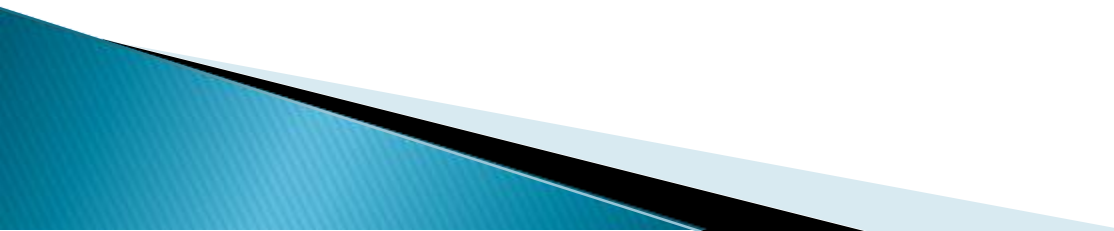


Diagnosis: Abscess



- ▶ A collection of pus within the dermis and deeper skin tissues
- ▶ Present as painful, tender, fluctuant and erythematous nodules
- ▶ Often surmounted by a pustule and surrounded by a rim of erythematous edema
- ▶ Spontaneous drainage of purulent material may occur

What is/are the next step(s) in management?

- a) Incision and drainage
 - b) Topical antibiotics
 - c) Offer HIV test
 - d) a and b
 - e) a and c
- 

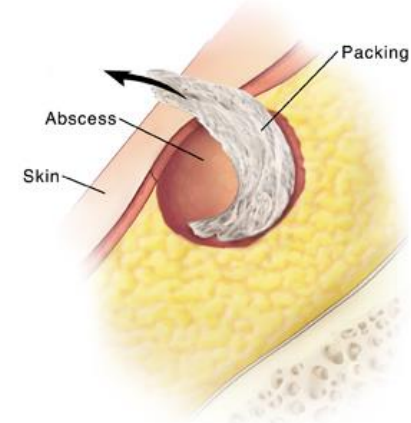
Answer...

- a) Incision and drainage
- b) Topical antibiotics (not effective)
- c) Offer HIV test
- d) a and b
- e) a and c correct



Abscess: Treatment

- ▶ Abscesses require incision and drainage (I & D)
 - Most experts recommend irrigation, breaking of loculations, and packing following incision and drainage



Abscess: Treatment (cont.)

- ▶ Recommend antibiotics in patients with severe/extensive disease, rapid progression or failure to improve with I&D, systemic illness, associated comorbidities, extremes of age
- ▶ Wound cultures should always be sent

Do you know the following diagnoses?

Hint: these skin infections involve hair follicles

What is the diagnosis?



Furunculosis

A furuncle (boil) is an acute, round, tender, circumscribed, perifollicular abscess that generally ends in central suppuration



What is the diagnosis?



Carbunculosis

- ▶ A carbuncle is a coalescence of several inflamed follicles into a single inflammatory mass with purulent drainage from multiple follicles





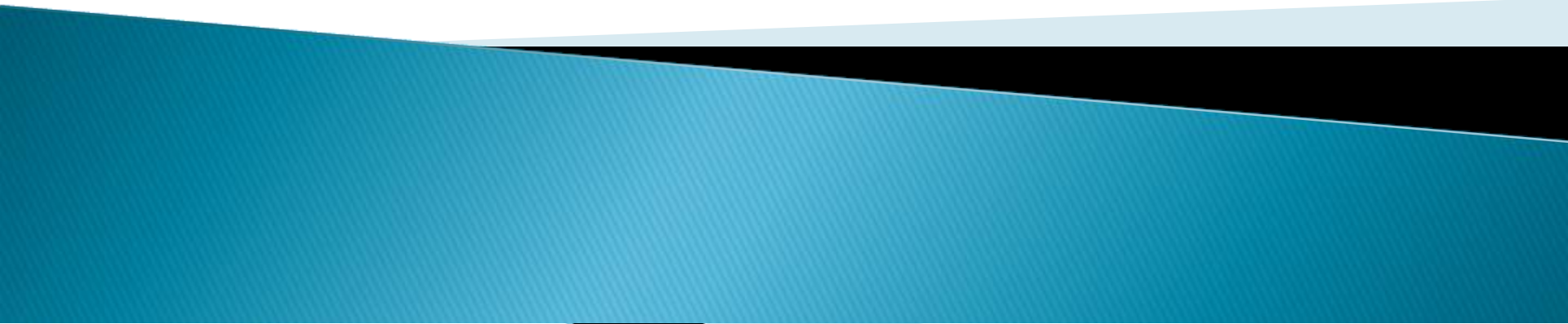
Furuncle, Carbuncle



- ▶ Abscess subtypes in areas where hair follicles are exposed to friction and perspiration
 - Neck, face, axillae, buttocks
- ▶ Usually **Staphylococcus aureus**
- ▶ Tx: oral antibiotics
 - Small solitary furuncle may drain wth just **warm compresses**
 - Larger furuncles and carbuncles: **treat like an abscess**

Jeffrey Anders

Case



History

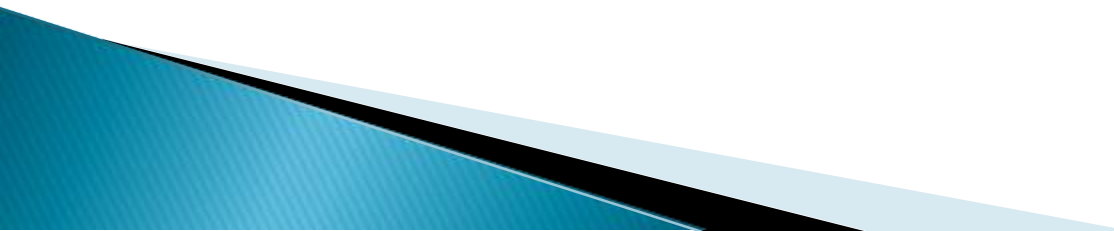
- ▶ Mr. Anders is a 19-year-old man who presents to dermatology clinic with two weeks of multiple “pimples” in his groin. He is concerned he has an STD.
- ▶ When asked, he reports occasionally shaving his pubic hair
- ▶ Sexual history reveals one female partner in the last year

Case Four: Skin Exam

- ▶ Multiple follicular pustules with surrounding erythema in the right groin



Which of the following recommendations would you provide Mr. Anders?

- a) Prescribe oral antibiotics
 - b) Stop shaving that area
 - c) Wash the area with antibacterial soap
 - d) All of the above
- 

Answer...

- a) Prescribe oral antibiotics
- b) Stop shaving that area
- c) Wash the area daily (antibacterial soap may be used)
- d) All of the above



Folliculitis

- ▶ Folliculitis is a superficial bacterial infection of the hair follicles
- ▶ Small, raised, erythematous, occasionally pruritic pustules <5 mm in diameter
- ▶ Pathogens:
 - Majority of cases are due to *Staphylococcus aureus*
 - If there has been exposure to a **hot tub** or swimming pool, consider **pseudomonas folliculitis** as a possible cause

Folliculitis: Management

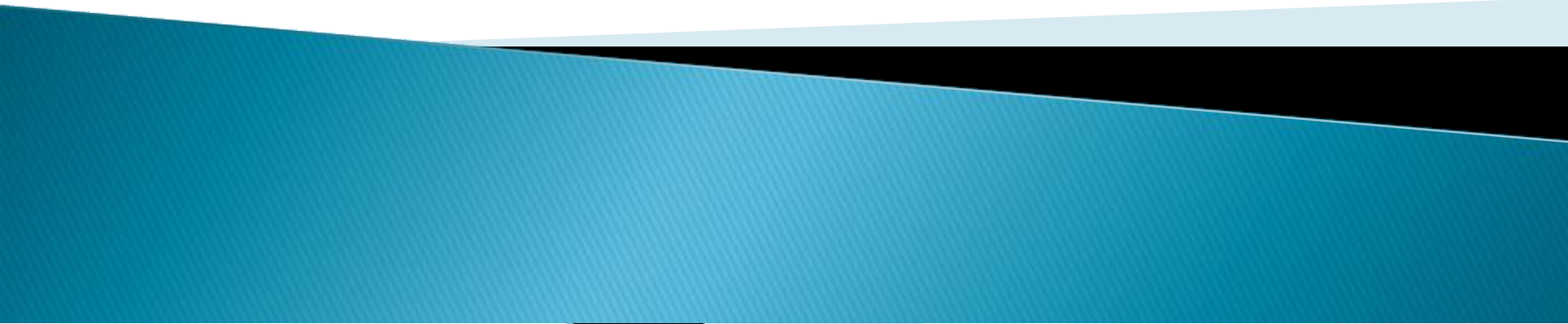
- ▶ Wash with antibacterial soap 3x/day
- ▶ Superficial pustules will rupture and drain spontaneously
- ▶ Oral or topical anti-staphylococcal agents may be used
- ▶ Deep lesions are small abscesses → drain them!

More Examples of Folliculitis



Danny Holden

Case



History

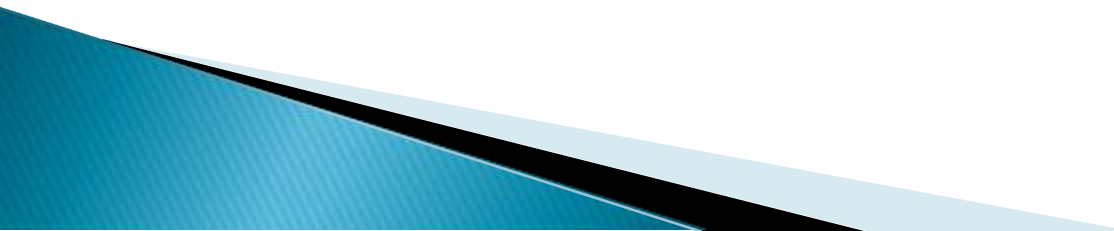
- ▶ Mr. Holden is a 20 year-old man who presents to his primary care provider with a three-week history of a facial rash. The rash is not painful, but occasionally burns and itches.
- ▶ He tried over the counter hydrocortisone cream with no relief.

Skin Exam

- ▶ Peri-oral papules and plaques with overlying honey-colored crust
- ▶ Minimal surrounding erythema



What is the most likely diagnosis?

- a) Acne vulgaris
 - b) Impetigo
 - c) Orolabial HSV
 - d) Seborrheic dermatitis
 - e) Perioral Dermatitis
- 

Answer...

- a) Acne vulgaris (would expect comedones and pustules, but not crusted plaques)
- b) **Impetigo**
- c) Orolabial HSV (would expect grouped and confluent vesicles with an erythematous rim; can evolve to crusting and easily be confused with impetigo)
- d) Seborrheic dermatitis (would expect erythematous patches and plaques with a greasy, yellow scale)
- e) Perioral Dermatitis (will present like Acne, but just around the mouth and is treated with antibiotics– usually Erythromycin)



Acne
Seb. Derm



HSV
Perioral Derm





Diagnosis: Impetigo

- ▶ Impetigo is a common superficial bacterial skin infection
- ▶ Most commonly seen in children ages 2–5
- ▶ Easily spread by close contact
- ▶ Usually *S. aureus*. Sometimes *S. pyogenes* or a combination of these two organisms
- ▶ 3 clinical variants (non-bullous impetigo, bullous impetigo, and ecthyma)

Examples of Non-bullous Impetigo

- ▶ Also called **impetigo contagiosum**
- ▶ Lesions begin as papules surrounded by erythema, progress to enlarging pustules that break down and form thick, adherent “golden” or “honey colored” crusts



Example of Bullous Impetigo

- A form of impetigo seen in young children is characterized by flaccid bullae with clear yellow fluid, which later becomes purulent.
- Ruptured bullae leave a thick brown crust



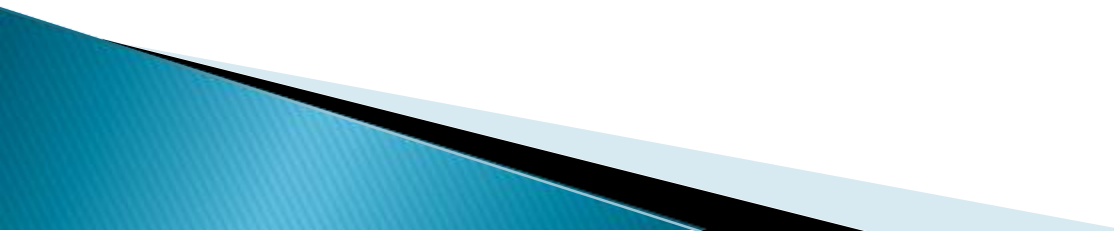
Ecthyma

- ▶ Ecthyma is an ulcerative form of impetigo in which the lesions extend through the epidermis and deep into the dermis.
- ▶ They consist of “punched out” ulcers covered with yellow crust surrounded by raised margins.



Danny Holdon was diagnosed with non-bullous impetigo based on clinical findings.

Which of the following treatment recommendations is/are most appropriate for Danny?

- a) Hand washing to reduce spread
 - b) Topical or oral antibiotics
 - c) Wash the affected area with antibacterial soap
 - d) All of the above
- 

Answer...

- a) Hand washing to reduce spread
- b) Topical or oral antibiotics
- c) Wash the affected area with antibacterial soap
- d) All of the above

Impetigo: Treatment

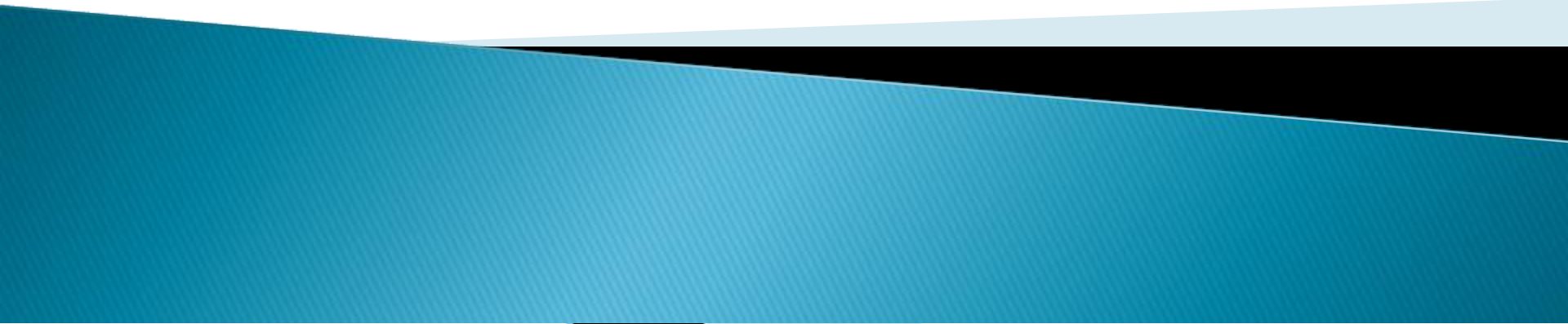
- ▶ Dicloxacillin
- ▶ Cephalexin
- ▶ Erythromycin (some strains of *Staphylococcus aureus* and *Streptococcal pyogenes* may be resistant)
- ▶ Clindamycin
- ▶ Amoxicillin/clavulanate
- ▶ Mupirocin ointment – may be equally effective to orals if lesions are localized in an otherwise healthy patient

Oral antibiotics

Topical therapy

Rodney Gorton

Case



History

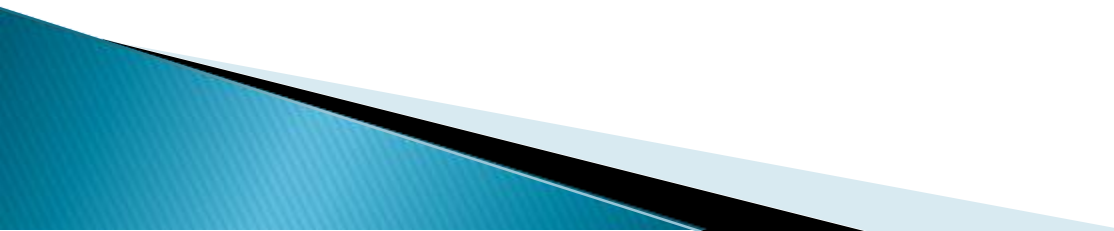
- ▶ HPI: Mr. Gorton is a 66-year-old man who was admitted for an inguinal hernia repair. His surgery went well and he was recovering without complication until he was found to have an expanding red rash on his left thigh. The dermatology service was consulted for evaluation of the rash.
- ▶ PMH: hypertension, diabetes mellitus type 2
- ▶ Medications: lisinopril, insulin, oxycodone
- ▶ Allergies: none
- ▶ Family history: noncontributory
- ▶ Social history: retired, lives with his wife
- ▶ Health-related behaviors: no alcohol, tobacco, or drug use
- ▶ ROS: febrile, fatigue, rash is painful

Exam

- Vital signs: T 101.1, HR 110, BP 90/50, RR 18, O2 sat 98%
- General: ill-appearing gentleman lying in bed
- Skin: ill-defined, large erythematous plaque with central patches of dusky blue discoloration, which is anesthetic, upon **re-examination 60 minutes later the redness had spread**



Which of the following do you recommend for initial management?

- a) Call an urgent surgery consult
 - b) Give IV fluids and antibiotics
 - c) Image with stat MRI
 - d) Obtain a deep skin biopsy
 - e) All of the above
- 

Answer...

- a) Call an urgent surgery consult (The suspected diagnosis is a surgical emergency)
- b) Give IV fluids and antibiotics (Patients quickly become hemodynamically unstable)
- c) Image with stat MRI (To assess degree of soft tissue involvement. Appropriate, but do not delay surgical intervention)
- d) Obtain a deep skin biopsy (Helps confirm diagnosis)
- e) All of the above

Diagnosis: Necrotizing Fasciitis

- ▶ Necrotizing fasciitis is a severe infection of tissue at the fascial level with 20% mortality rate
- ▶ Progresses rapidly (hours) following surgery, trauma, or nothing at all
- ▶ Expanding dusky, edematous, red/blue/purple area, possibly with crepitus, anesthesia
- ▶ Usually polymicrobial: **group A streptococcus (GAS)** **most common**, Staphylococcus aureus, Clostridium, etc
- ▶ Systemic symptoms: fever, tachycardia, diarrhea, vomiting

Necrotizing Fasciitis: Treatment



1. Wide surgical debridement
 2. Broad-spectrum systemic antibiotics
-
- ▶ Poor prognostic factors:
 - Delay in diagnosis
 - Age > 50
 - Diabetes
 - Atherosclerosis
 - Infection involving the trunk

Take Home Points



- ▶ **Cellulitis** is a bacterial infection of the dermis that often begins with a portal of entry that is usually a wound, insect bite, or fungal infection (tinea pedis)
- ▶ It is important to recognize and treat cellulitis early as untreated cellulitis may lead to sepsis and death
- ▶ **Erysipelas** is a superficial cellulitis with marked dermal lymphatic involvement
- ▶ A skin **abscess** is a loculated infection within the dermis and deeper skin tissues and is best treated with I&D
- ▶ **Furuncles** and **carbuncles** are a subtype of abscesses, which preferentially occur in skin areas containing hair follicles exposed to friction and perspiration

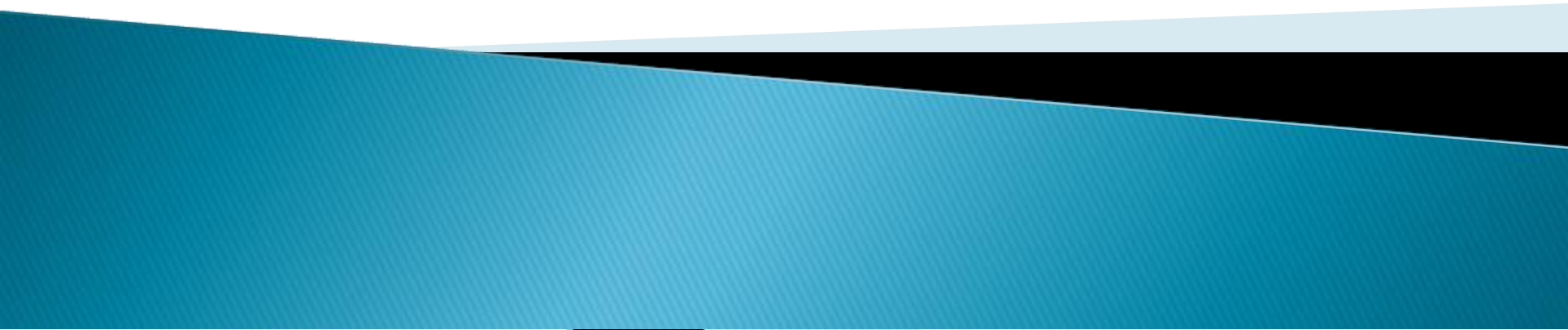
Take Home Points (cont.)



- ▶ **Folliculitis** is a superficial bacterial infection of the hair follicles presenting as follicular pustules
- ▶ In **impetigo**, papules and vesicles progress to form pustules that enlarge and break down to form thick, adherent crusts with a golden or honey-colored appearance
- ▶ **Necrotizing fasciitis** presents as an expanding dusky, edematous, red plaque with blue discoloration
 - Anesthesia of the skin of the affected area is a characteristic finding
 - medical/surgical emergency

Michael White

Case



History

- ▶ HPI: Michael White is a healthy 54-year-old who presents with thickening of several toenails that did not improve with topical miconazole cream nor an over-the-counter mentholated vapor rub.
- ▶ PMH: no major illnesses or hospitalizations
- ▶ Medications: atorvastatin
- ▶ Allergies: none
- ▶ Family history: mother with psoriasis
- ▶ Social history: Retired firefighter and part-time swimming coach. Lives with his second wife.

Skin Exam

- ▶ How would you describe these exam findings?

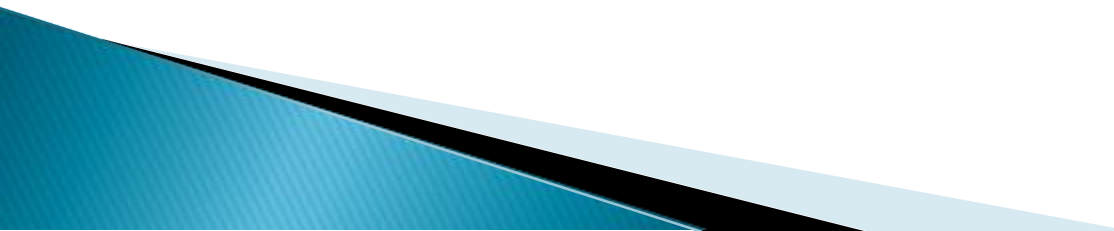


Skin Exam

- ▶ Nail thickening and subungual debris
- ▶ Additional exam
 - Multiple toenails involved
 - Also scaling on soles bilaterally
 - Rest of full body skin exam is normal



What is the best test to order at this time?

- a) Clinical diagnosis is sufficient
 - b) Fungal culture of subungual debris
 - c) Hepatitis C antibodies
 - d) HIV 1 and 2 antibodies
 - e) Punch biopsy of nail plate and bed
- 

Answer



- a) Clinical diagnosis is sufficient (low specificity; often not fungus; need confirmation of fungus!)
- b) **Fungal culture of subungual debris**
- c) Hepatitis C antibodies (associated with lichen planus, not onychomycosis)
- d) HIV 1 and 2 antibodies (not indicated for distal subungual onychomycosis)
- e) Punch biopsy of nail plate and bed (too invasive for first test; fungal culture is better)



Onychomycosis

- ▶ Chronic fungal infection of the nailbed
 - Usually starts with tinea pedis
- ▶ Responds very poorly to topical antifungals
- ▶ **Confirmation fungus before oral treatment**
 - May mimic other conditions (e.g. psoriasis)
 - Fungal culture is preferred to direct therapy

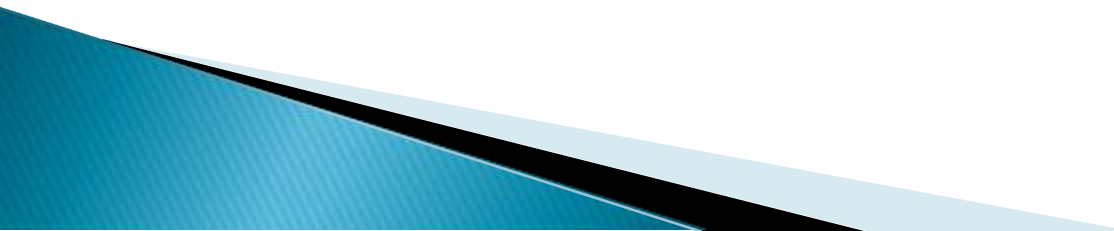


Distal subungual onychomycosis

- ▶ Thickened nail, subungual debris, and separation of the nail plate from nail bed
- ▶ Usually caused by dermatophyte *Trichophyton rubrum*



The fungal culture comes back three weeks later, positive for *Trichophyton rubrum*. What is the best treatment?

- a) Ciclopirox nail lacquer
 - b) Clotrimazole cream
 - c) Oral mupirocin
 - d) Oral nystatin
 - e) Oral terbinafine
- 

Answer...

- a) Ciclopirox nail lacquer (poor efficacy compared to oral antifungals)
- b) Clotrimazole cream (poor efficacy)
- c) Oral mupirocin is an antibiotic, not an antifungal
- d) Oral nystatin (only works on Candida yeasts)
- e) **Oral terbinafine**



When to refer



This patient has lichen planus, not onychomycosis

- ▶ Negative fungal cultures
 - Only half of nail dystrophy is caused by fungus
- ▶ Atypical pattern
- ▶ Other rashes are present
- ▶ Fails first-line therapy

Summary: Adult Fungal Treatment

	1 st line	2 nd line
Tinea pedis Tinea corporis Tinea cruris	Clotrimazole Miconazole cream	Terbinafine Naftifene Butenafine cream or gel
Tinea versicolor	Selenium sulfide, zinc pyrithione shampoos	Oral fluconazole or itraconazole
Candida	Clotrimazole Miconazole cream or Nystatin powder, ointment	Desonide or 1% hydrocortisone ointment
Seborrheic dermatitis	Ketoconazole cream	Desonide or 1% hydrocortisone for flares

Take Home Points



- ▶ Cutaneous fungal infections are extremely common.
- ▶ If it scales, scrape it! KOH examination is the easiest and most cost effective method used to diagnose fungal infections of the hair, skin, and nails.
- ▶ Fungal culture is helpful in tinea corporis when the source of infection is not obvious (as opposed to tinea pedis)

Take Home Points



- ▶ Topical treatment is usually appropriate as a first-line agent for tinea pedis, tinea corporis, and candidal intertrigo, however oral medications are called for when involvement is extensive, when tinea corporis is thought to have been transmitted by an animal, and in fungal infections of the nails.
- ▶ Positive fungal culture is recommended prior to initiating oral antifungal treatment for onychomycosis
- ▶ Monitor liver function tests when starting oral antifungals
- ▶ Fungal infections have high rates of recurrence after treatment, but maintaining a dry, clean skin environment is helpful for prevention.

Anna Jones

Case

History

- ▶ HPI: Ms. Jones is a 27-year-old woman who presents with mild itchiness of her back which began mid summer. She is also concerned about areas on her back that are darker than the surrounding skin.
- ▶ PMH: asthma
- ▶ Medications: occasional multivitamin
- ▶ Allergies: no known drug allergies
- ▶ Social history: spends her summer months in Florida and is an avid runner.

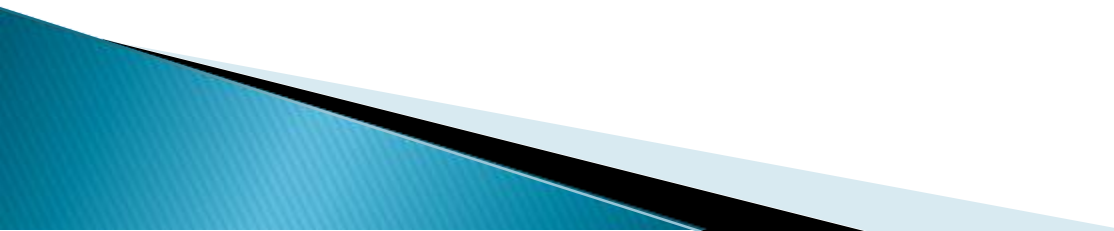
Skin Exam



How would you describe these exam findings?

Well-demarcated, hyperpigmented macules and patches, across the back.

Which of the following is the most likely diagnosis?

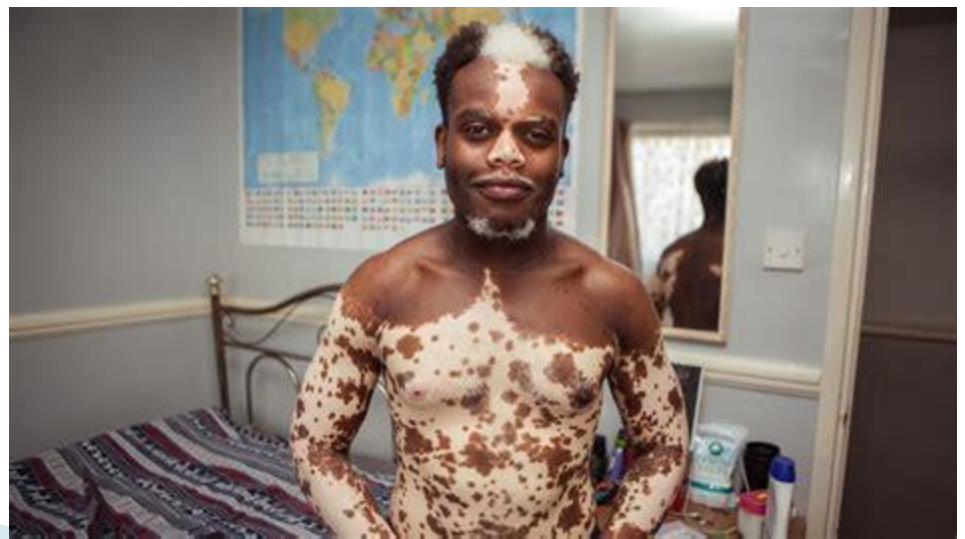
- a) Pityriasis alba
 - b) Seborrheic keratosis
 - c) Tinea corporis
 - d) Tinea versicolor
 - e) Vitiligo
- 



Pityriasis alba
Seborrheic
dermatitis



Tinea corporis
Vitiligo



Answer...

- a) Pityriasis alba (noninfectious, asymptomatic poorly-defined areas of hypopigmentation; self-limited)
- b) Seborrheic dermatitis (abnormal immune response to normal skin yeast causing scaling and crusting)
- c) Tinea corporis (fungal skin infection, presents as erythematous annular lesions with central clearing)
- d) **Tinea versicolor**
- e) Vitiligo (autoimmune loss/dysfunction of melanocytes causing areas of complete depigmentation)

Diagnosis: Tinea Versicolor



- ▶ Tinea versicolor (aka Pityriasis versicolor) is not caused by a dermatophyte
- ▶ It is a colonization caused by species of *Malassezia*, a lipophilic **yeast** that is a normal resident in the keratin of the skin and hair follicles of individuals at puberty and beyond
- ▶ Tends to recur annually in the summer months

Tinea Versicolor

- Characterized by well-demarcated, tan, salmon, or hypopigmented or hyperpigmented patches, occurring most commonly on the trunk and arms
- Macules will grow, coalesce and various shapes and sizes are attained in an asymmetric distribution
- Visible scale is not often present, but when rubbed with a finger or scalpel blade, scale is readily seen
 - This is a diagnostic feature of tinea versicolor
 - Evoked scale will disappear after treatment

Tinea Versicolor: Morphology



“Versi”tile appearance:

On light skin → salmon-colored or brown lesions.

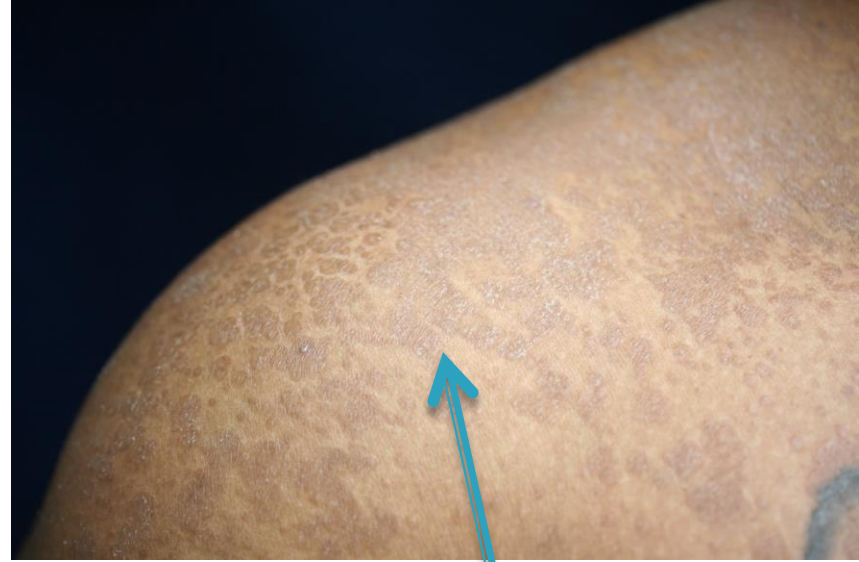
On tan skin → pale in comparison.

On darker skin → hyper- or hypopigmented lesions.

Examples of Tinea Versicolor



Light spots

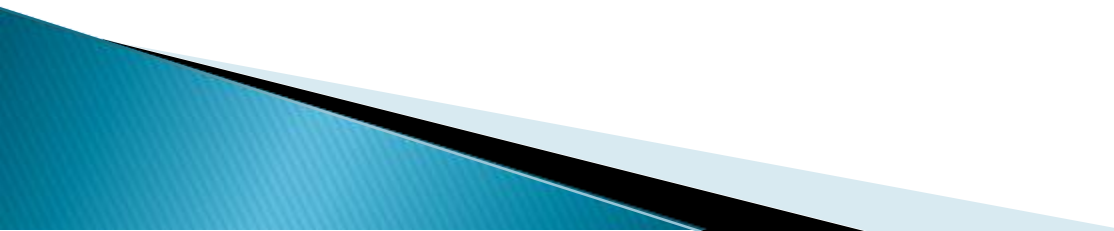


Dark spots

Pink or tan
spots



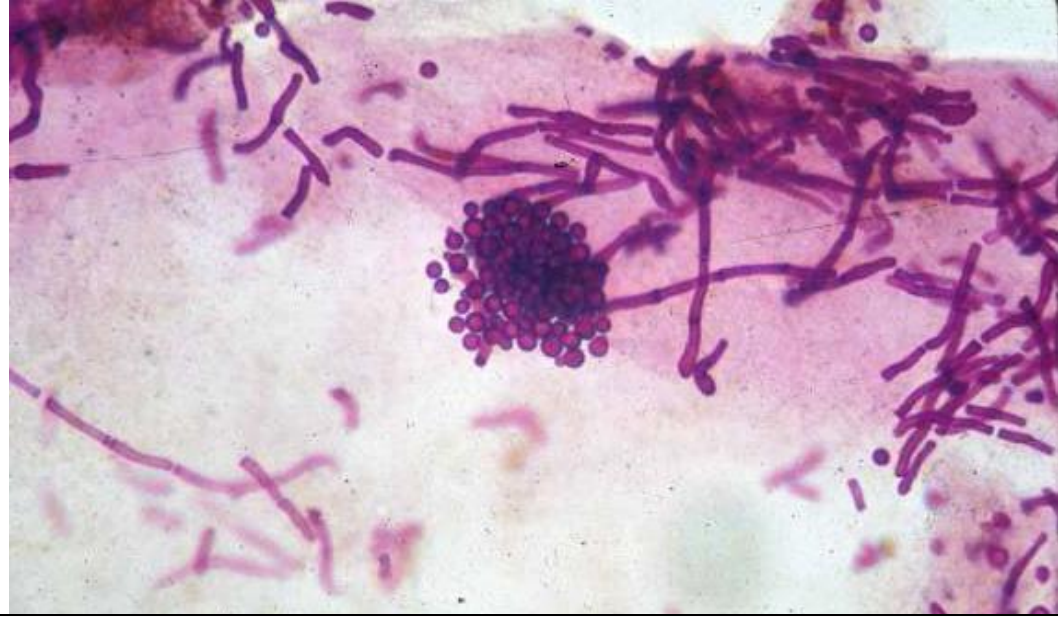
Which of the following is the most appropriate next step in management?

- a) Fungal culture
 - b) KOH exam
 - c) Skin biopsy
 - d) Wood's light exam
- 

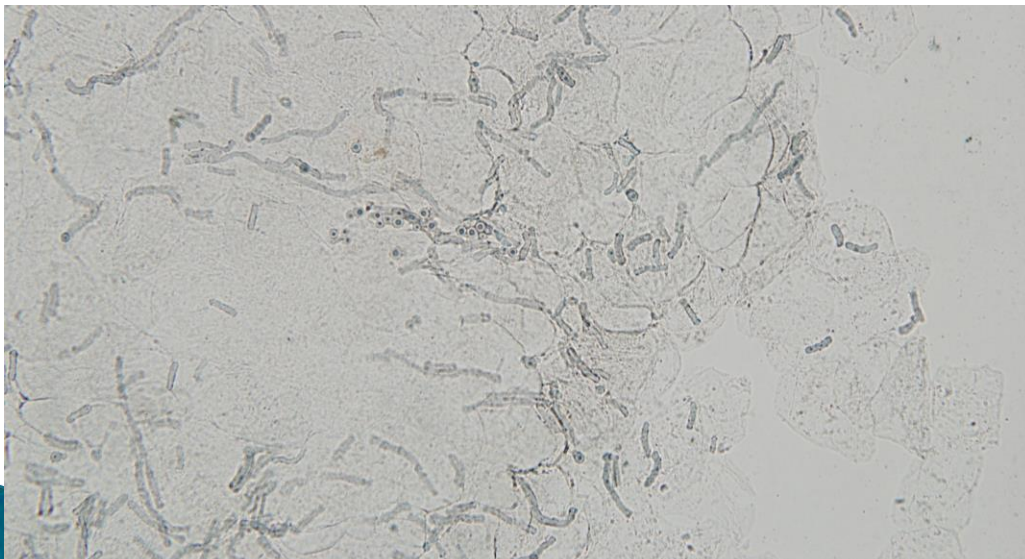
Answer...

- a) Fungal culture (*Malassezia* spp. are easily identified by a KOH exam but are not easily cultured)
- b) **KOH exam**
- c) Skin biopsy (too invasive; KOH is better)
- d) Wood's light exam (mild fluorescence may be present but not as sensitive as KOH exam)

Spaghetti and Meatballs on Microscopy

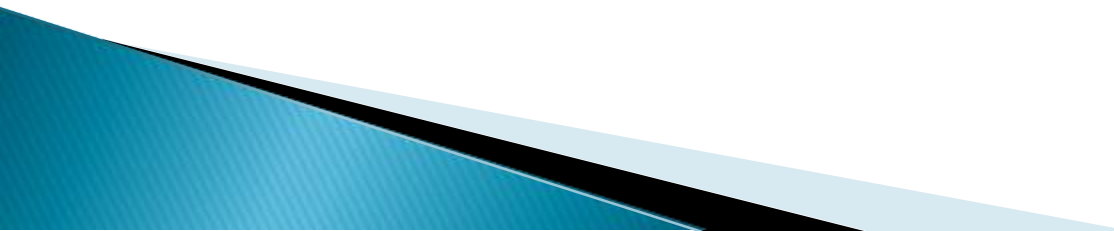


Microscopy with dye added to the specimen



The KOH exam shows short hyphae and small round spores.

Which of the following treatments would you recommend for Ms. Jones?

- a) Antifungal shampoo
 - b) Hydroquinone cream
 - c) Nystatin cream
 - d) Oral terbinafine
- 

Answer...

- a) Antifungal shampoo
- b) Hydroquinone cream (this bleaching cream is not an appropriate treatment)
- c) Nystatin cream (not effective)
- d) Oral terbinafine (in contrast to tinea corporis, this is not very effective in tinea versicolor)

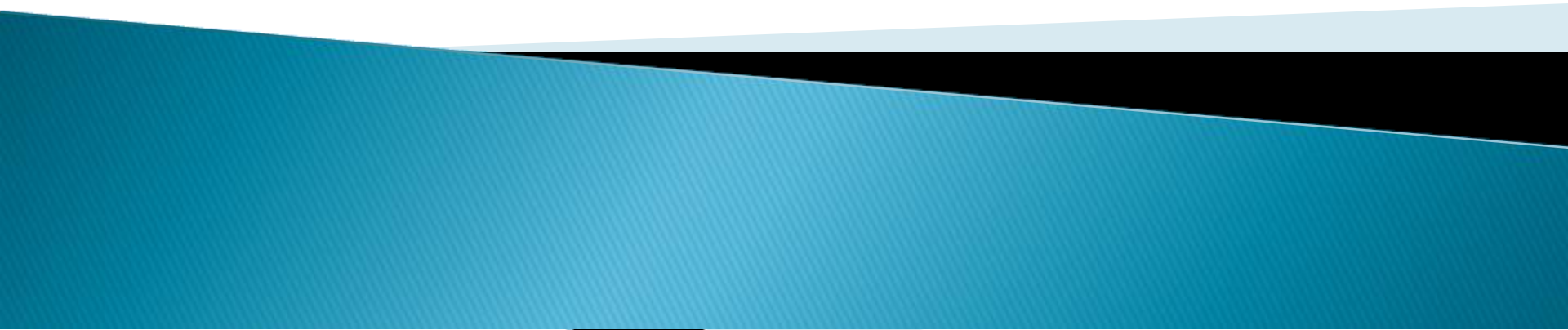
Topical Treatment is First-line



1. Shampoos: selenium sulfide 2% shampoo, ketoconazole shampoo, zinc pyrithione shampoo
 2. Imidazole creams: ketoconazole, clotrimazole
- Use orals when topical therapy fails or the patient has a strong preference for oral therapy
 - Fluconazole or Itraconazole
 - Monitor liver function if giving more than 7 days
 - Many patients relapse! Maintenance with shampoos.

John Wong

Case



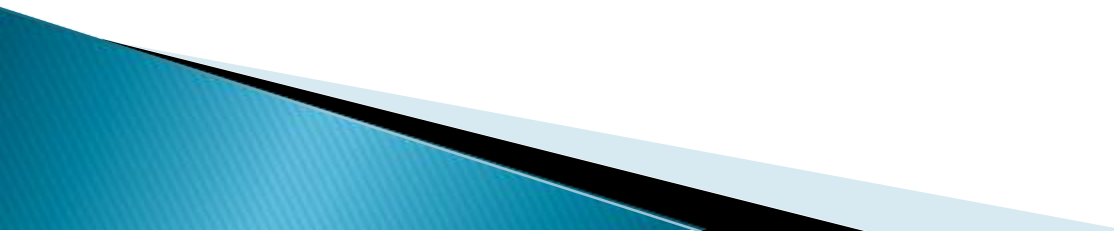
History

- ▶ HPI: Mr. Wong is a 59 year-old man with several years of redness and scaling on his forehead, eyebrows, and central face. It improves slightly with moisturizers and does not itch. It does not worsen with heat, exercise, or alcohol.
- ▶ PMH: none
- ▶ Allergies: none
- ▶ Medications: allopurinol
- ▶ Family History: non-contributory
- ▶ Social History: accountant

Skin Exam



How would you describe the rash on Mr. Wong's face?

- a) Erythematous patches with overlying scale
 - b) Lichenified plaques with silvery scale
 - c) Papules and pustules
 - d) Vesicles and crust
- 

Answer...

- a) Erythematous patches with overlying scale
- b) Lichenified plaques with silvery scale
- c) Papules and pustules
- d) Vesicles and crust



What is the most likely diagnosis?

- a) Actinic keratoses
- b) Allergic contact dermatitis
- c) Atopic dermatitis
- d) Rosacea
- e) Seborrheic dermatitis



Answer...

- a) Actinic keratoses (scaly is greasy, not keratotic)
- b) Allergic contact dermatitis (he does not itch)
- c) Atopic dermatitis (very itchy; no history of this)
- d) Rosacea (no history of typical triggers; usually telangiectasias or papules, not scaling)
- e) **Seborrheic dermatitis**





Actinic keratoses
Allergic contact
dermatitis



Atopic dermatitis
Rosacea



Seborrheic dermatitis

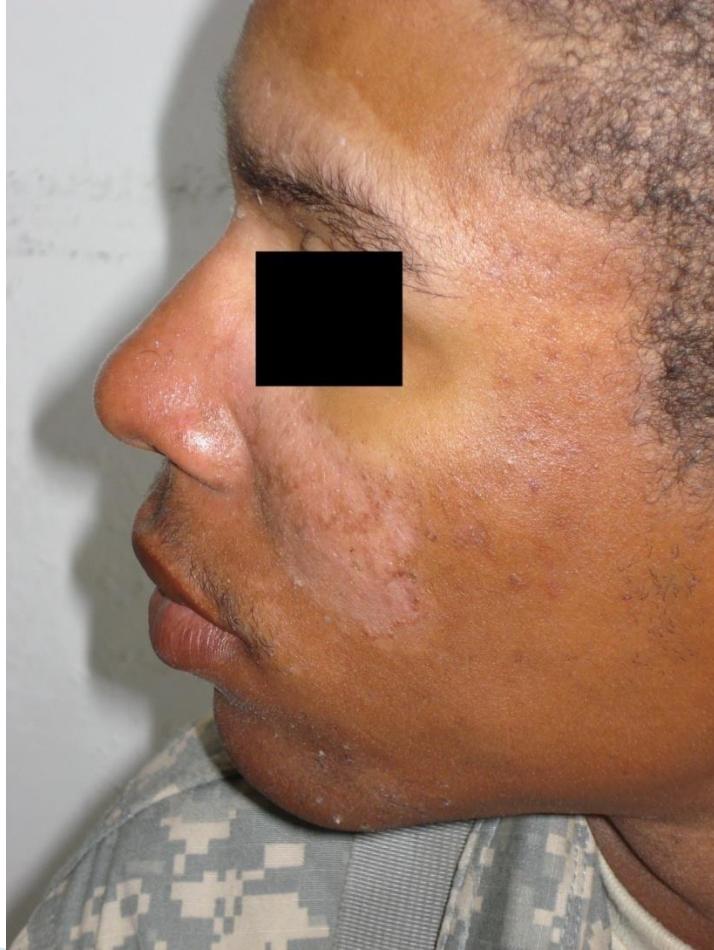


- ▶ Seborrheic dermatitis is a very common inflammatory reaction to the *Malassezia* (*Pityrosporum ovale*) yeast that thrives on seborrheic (oil-producing) skin
 - Inflammatory reaction to normal flora
 - This is a chronic condition than can be controlled but not cured
- ▶ Presents as erythematous scaling patches on the scalp, hairline, eyebrows, eyelids, central face and nasolabial folds, external auditory canals, or central chest

More examples of Seborrheic dermatitis



Seborrheic dermatitis



Often hypopigmented
in darker skin types

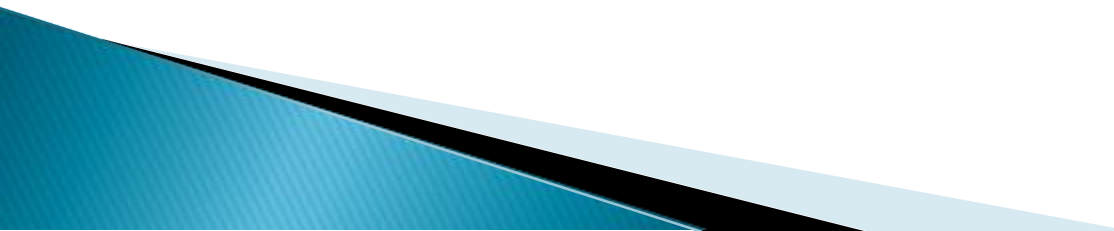
Seborrheic dermatitis



- Favors central chest
- May be erythematous, hypopigmented
- If partially treated, may be just a few papules



Which of the following would be an appropriate treatment for Mr. Wong?

- a) Clobetasol propionate cream
 - b) Desonide cream (mild steroid, low potency)
 - c) Erythromycin ointment
 - d) Nystatin cream
 - e) 5-fluorouracil cream
- 

Answer...

- a) Clobetasol proprionate cream (too potent)
- b) Desonide cream (mild steroid, low potency)
- c) Erythromycin ointment (this is not bacterial)
- d) Nystatin cream (for Candida yeast species, not Malassezia species)
- e) 5-fluorouracil cream (for actinic keratoses)



Seborrheic dermatitis treatment



- ▶ Topical ketoconazole twice daily
 - Reduces yeast counts; choose vehicle based on patient
- ▶ Low-potency topical steroid (e.g. desonide) are safe to use for flares on the face
 - Use twice daily for 1–2 weeks for flares
- ▶ Antidandruff shampoo for scalp, chest
 - Ketoconazole, selenium sulfide, zinc pyrithione shampoos
 - Lather, leave on 10 minutes, rinse; repeat 3–5x/week
- ▶ Refer patients who fail these therapies

Summary: Adult Fungal Treatment

	1 st line	2 nd line
Tinea pedis Tinea corporis Tinea cruris	Clotrimazole Miconazole cream	Terbinafine Naftifene Butenafine cream or gel
Tinea versicolor	Selenium sulfide, zinc pyrithione shampoos	Oral fluconazole or itraconazole
Candida	Clotrimazole Miconazole cream or Nystatin powder, ointment	Desonide or 1% hydrocortisone ointment
Seborrheic dermatitis	Ketoconazole cream	Desonide or 1% hydrocortisone for flares

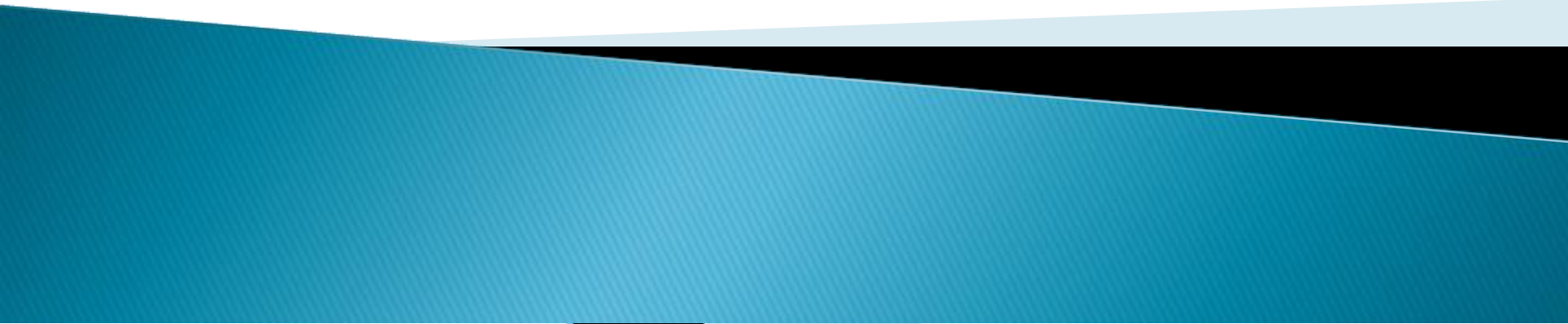
Take Home Points



- ▶ Tinea versicolor is characterized by well-demarcated, tan, salmon, or hypopigmented patches, occurring most commonly on the trunk. KOH confirms the diagnosis.
- ▶ Candida intertrigo classically has satellite macules and papules or pustules around patches of erythema
- ▶ Topical treatment is usually appropriate as a first-line agent for tinea versicolor, candidal intertrigo, and seborrheic dermatitis
- ▶ Tinea versicolor and seborrheic dermatitis are chronic conditions requiring maintenance therapy
- ▶ Monitoring for recurrence and maintenance treatments may be helpful in patients with recurrent infection

Megan Driskell

Case



History

- ▶ HPI: Megan is an 8-year-old girl who presents to her pediatrician's office with bumps on her fingers and hands. They have been present for 3 months without change and are asymptomatic.
- ▶ PMH: no chronic illnesses or prior hospitalizations
- ▶ Allergies: no known allergies
- ▶ Medications: none
- ▶ Family history: no affected family members
- ▶ Social history: lives at home with parents and attends school
- ▶ ROS: negative

Skin Exam



How would you describe these lesions?



- a) Hyperkeratotic and umbilicated papules and nodules
- b) Hyperkeratotic, endophytic papules and nodules
- c) Hyperkeratotic, exophytic papules and nodules
- d) Smooth and umbilicated papules and nodules
- e) Smooth, exophytic papules and nodules

Answer...

- a) Hyperkeratotic and umbilicated papules and nodules (these papules are not umbilicated)
- b) Hyperkeratotic, endophytic papules and nodules (these papules are growing outward, not inward)
- c) **Hyperkeratotic, exophytic papules and nodules**
- d) Smooth and umbilicated (marked by a depressed spot) papules and nodules (more characteristic of molluscum contagiosum)
- e) Smooth, exophytic papules and nodules (these papules are not smooth)

Clinical Features of Verruca Vulgaris



- ▶ Hyperkeratotic, exophytic, dome-shaped papules or nodules
- ▶ Most common on fingers, dorsal hands, knees or elbows but may occur anywhere
- ▶ Punctate black dots represent thrombosed capillaries
- ▶ May **koebnerize** – spread with skin trauma



Verruca vulgaris is caused by:

- a) Human immunodeficiency virus
- b) Human papillomavirus
- c) Pox virus
- d) Herpes virus
- e) Varicella-zoster virus

Answer...

- a) Human immunodeficiency virus
- b) **Human papillomavirus**
- c) Pox virus
- d) Herpes virus
- e) Varicella–zoster virus

Human Papillomavirus (HPV): Overview



- HPV infects basal keratinocytes of skin and mucosal epithelia, causing hyperplasia of the epithelium → wart
- Transmitted by skin-to-skin contact or contaminated surfaces or objects

Role of HPV in Cutaneous Disease

- ▶ The type of HPV influences the wart morphology

Type of Wart	HPV Type
Verruca vulgaris: common warts	HPV 2,4
Verrucae plana: flat warts	HPV 3, 10
Palmoplantar warts	HPV 1
Condylomata acuminata: external genital warts	HPV 6, 11, 16, 18, 31, and more

Verrucae Planae: Flat Warts



- ▶ Skin-colored or pink
- ▶ Smooth-surfaced, slightly elevated, flat-topped papules
- ▶ Dorsal hands, arms, face (exposed surfaces)

Palmoplantar Verruca

- ▶ Thick, endophytic papules
- ▶ Plantar warts may be painful when walking



General Treatment Principles



- ▶ Observation: Chance of spontaneous resolution at 2 years is over 75%
- ▶ Multiple treatments are almost always needed for any treatment modality (laser, acids, cryotherapy, etc.)
- ▶ Start with therapies that minimize side effects and risk of scarring

Common Wart Treatment Options

- ▶ Treatments are often destructive and/or stimulate the immune system
 - There is no specific anti-HPV therapy (unlike HIV and viral hepatitis)

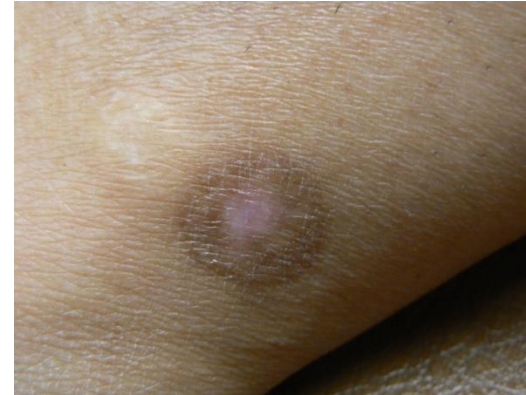
Primarily Irritating or Destructive	Primarily Immune-stimulating
Cryotherapy*	Imiquimod
Salicylic acid*	Candida antigen (IL)
Tretinoin cream	Immunotherapy: Squaric acid or DCNB
Shave removal	
Adhesive tape	
Laser	

Cryotherapy side effects:

- ▶ PIPA (post-inflammatory pigment alteration)
- ▶ Blister
- ▶ Pain
- ▶ Scar
- ▶ Nail dystrophy

Pre-

Post-



Megan and her mom have done some reading online about wart treatments. They ask about several treatments. Which of their suggestions below do you suggest **against**?

- a) cryotherapy
- b) excision
- c) imiquimod
- d) squaric acid



Which of their suggestions do you suggest
against?

- a) cryotherapy
- b) **Excision**
 - a) Least preferred therapy for this child with multiple warts on her hands. Excision is least favored because it is invasive, will leave scars and often does not completely eliminate HPV virus
- c) Imiquimod
 - a) Imiquimod is often less effective for this site but is not invasive like surgery
- d) squaric acid Cryotherapy, Salicyclic acid, and Squaric acid and DCNB are more favored (bold print above)

What is your favored therapy for this 6-year-old with flat warts on his face?

- a) Imiquimod
- b) Salicylic acid
- c) Shave removal
- d) Topical tretinoin



Answer...

- a) Imiquimod – not preferred because it can cause significant inflammation with erythema, crusting, and irritation
- b) Salicylic acid – not preferred because it can also cause significant inflammation and is difficult to cover on the face
- c) Shave removal – not preferred because of the risk of scarring as well as recurrence
- d) Tretinoin cream – the preferred treatment for this child with multiple warts on his face

Number of Warts	Hand & Foot	Face	Other Sites
Few	Cryotherapy Salicylic acid Adhesive tape Laser	Cryotherapy Excision/shave removal	Cryotherapy Salicylic acid
Many	Salicylic acid Cryotherapy Squaric acid or DCNB Laser	Cryotherapy Tretinoin cream	Imiquimod Tretinoin cream Squaric acid or DCNB

This 32 year old woman has a single wart on her foot. What is your preferred approach to cryotherapy?

- a) Freeze for 10 seconds, one time
- b) Freeze for 10 seconds, two times
- c) Following debridement of the keratotic tissue with emery board, pumice or sharply with scalpel, freeze with liquid nitrogen until the tissue visibly blanches, rewarms and repeat at least twice.



This 32 year old woman has a single wart on her foot. What is your preferred approach to cryotherapy?

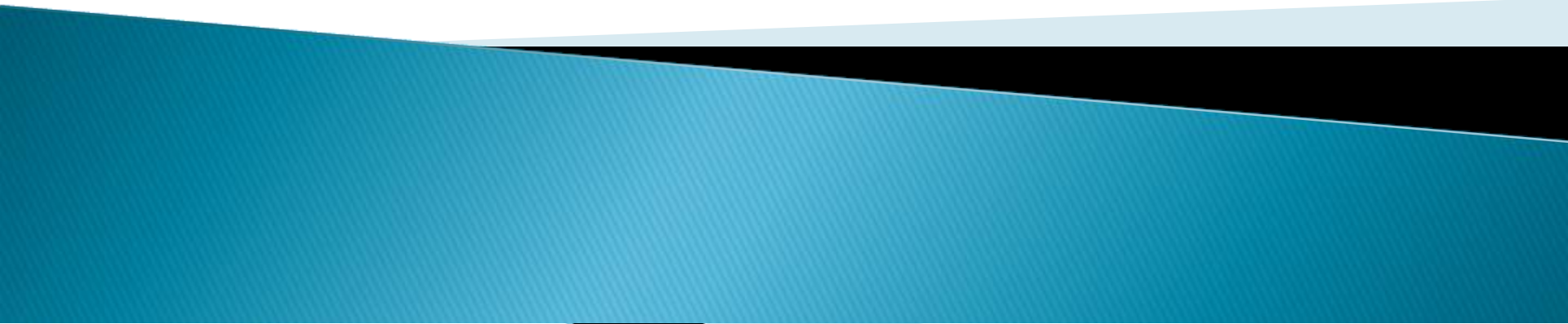
- * Following debridement of the keratotic tissue with emery board, pumice or sharply with scalpel, freeze with liquid nitrogen until the tissue visibly blanches, rewarms and repeat at least twice.

- Treatment is more effective if the thick overlying callus is removed – either sharply at the office or intermittently by the patient with a pumice stone at home.



Jonathan Cohen

Case

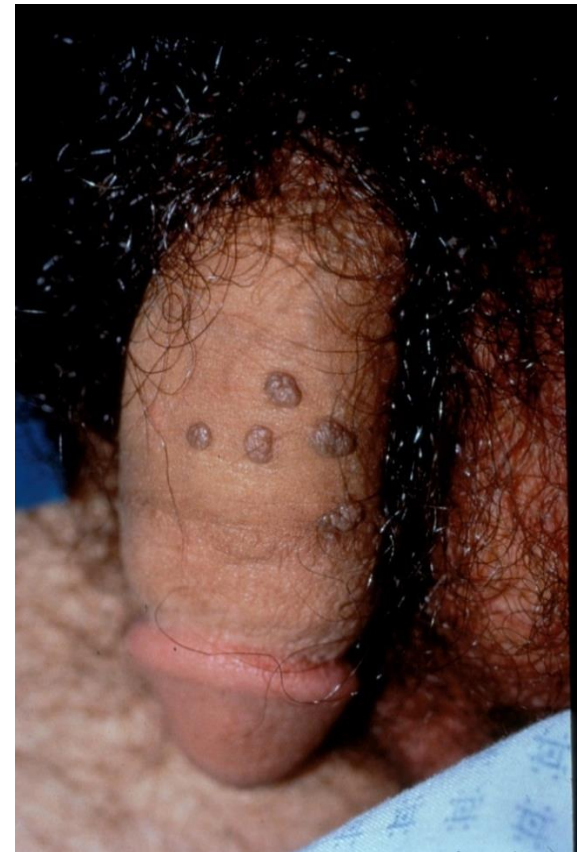


History

- ▶ HPI: Mr. Cohen is a 21-year-old man who comes into the STD clinic because of an increasing number of bumps on his penis over the last year.
- ▶ PMH: no chronic illnesses or prior hospitalizations
- ▶ Allergies: no known allergies
- ▶ Medications: none
- ▶ Family history: noncontributory
- ▶ Social history: studying economics at a nearby university
- ▶ ROS: negative

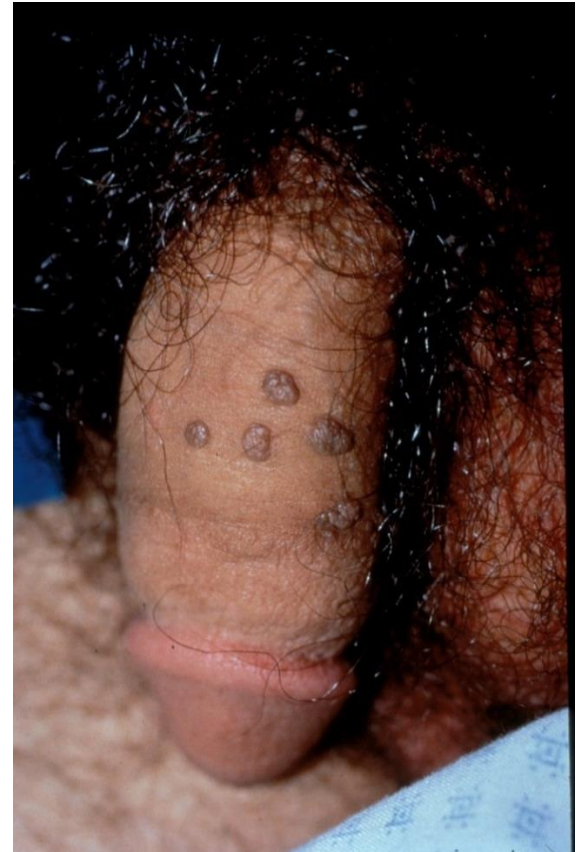
How would you describe these lesions?

- a) Pearly, vesicular papules
- b) Smooth, indurated plaques
- c) Soft, sessile papules
- d) Waxy, stuck-on plaques



How would you describe these lesions?

- a) Pearly, vesicular papules
- b) Smooth, indurated plaques
- c) **Soft, sessile papules**
- d) Waxy, stuck-on plaques



Clinical Features of external genital warts



- ▶ Soft sessile (broad based) papules
- ▶ Papules or large confluent plaques
- ▶ External genitalia, perineum, perianal, inguinal fold

Comparing: molluscum and genital warts

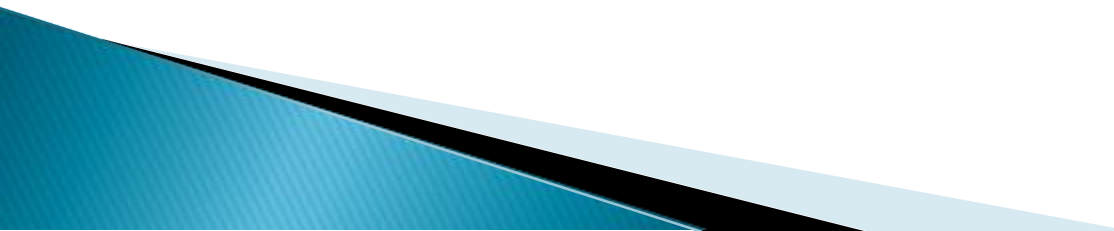


Molluscum Contagiosum
(smooth, dome-shaped papules with
central umbilication)



External Genital Warts
(hyperkeratotic, exophytic
papules and plaques)

What additional information is especially important to care for this patient?

- a) Medications
 - b) Sexual history
 - c) Surgical history
 - d) Allergies
- 

What additional information is especially important to care for this patient?

a) Medications

b) Sexual history

- to determine if the patient is at risk for anal HPV infection
- to discuss the contagious nature and risk of spreading to others and receipt from others
- to consider symptoms of additional concomitant STI

c) Surgical history

d) Allergies

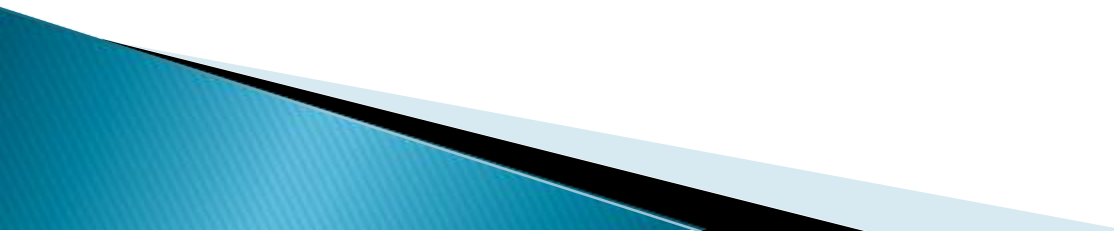


External Genital Warts (EGW)



- ▶ HPV infection is one of the most common STIs
 - Peak prevalence: women 20–24yo, men 25–29yo
 - Risk factors: sexual intercourse at an early age, numerous partners, unprotected exposure
- ▶ Effective prophylactic HPV vaccine is available for prevention of genital warts
 - Gardasil™ and Cervarix™
 - Both vaccines are effective against HPV types 16 and 18, which cause most cervical cancers

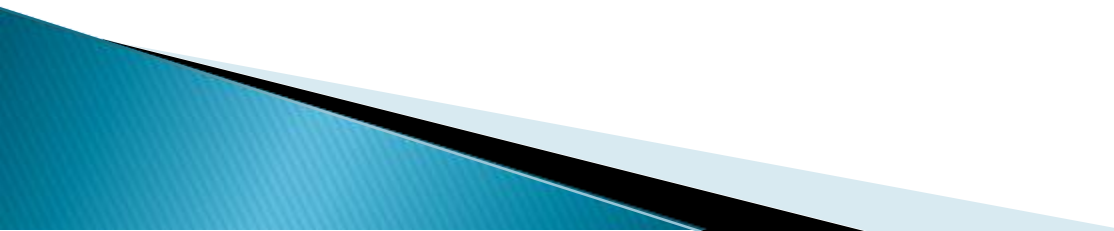
Which of the following treatments could you use for external genital warts?

- a) Cryotherapy
 - b) Imiquimod
 - c) Podophyllin
 - d) All of the above
- 

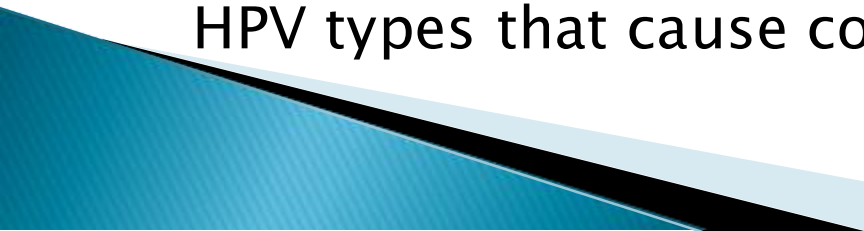
Answer...

- a) Cryotherapy (liquid nitrogen is used to freeze and destroy the infected tissue)
- b) Imiquimod (interferon inducer is topically applied 3x per week for 16 weeks)
- c) Podophyllin (this anti-mitotic agent is topically applied, twice daily for 3 days, wait 4 days, then repeat the cycle)
- d) All of the above

Parker is a 13-year old girl with warts on her fingers. She and her mother have heard that there is a vaccine that can treat warts and they want Parker to receive it. What is the best response to this request?

- a) She's too young to get the vaccine.
 - b) The types of HPV that the vaccine protects against are different from the HPV serotypes that are likely causing the warts on her hands.
 - c) The vaccine will help treat her warts and prevent more warts in the future.
- 

Answer...

- a) No, because she's too young to get the vaccine.
 - b) No, because the types of warts (HPV types) are different.
 - c) Yes, it will help treat and prevent warts.
-
- ▶ The HPV vaccines are FDA-approved for males and females 10–25 years old (option a) for prevention, not treatment (option c) of external genital warts and HPV-associated genital cancers, not common warts on the hands, face, feet, etc.
 - ▶ The FDA-approved vaccines are not composed of the HPV types that cause common warts
- 

Take Home Points



- ▶ Warts are caused by human papilloma viruses
- ▶ Numerous morphologies exist: common, flat, palmoplantar, external genital
- ▶ Treatment is difficult and there are many options available
- ▶ Vaccines have been developed to reduce HPV-associated neoplasia, and some have efficacy for external genital warts